MATH1014 Calculus II Tutorial
Week 7

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1 Review

1.1 Numerical Integration

When we cannot evaluate an integral analytically, we use numerical approximations. For an integral $\int_a^b f(x)\,dx$, divide the interval $[a,b]$ into n subintervals of equal width $\Delta x = \frac{b-a}{n}$. Let $x_i = a + i\Delta x$ for $i = 0, 1, \ldots, n$.

1. **Left Endpoint Approximation (L_n)**: Uses the left endpoint of each subinterval.

$$L_n = \Delta x[f(x_0) + f(x_1) + \cdots + f(x_{n-1})]$$

2. **Right Endpoint Approximation (R_n)**: Uses the right endpoint of each subinterval.

$$R_n = \Delta x[f(x_1) + f(x_2) + \cdots + f(x_n)]$$

3. **Trapezoidal Rule (T_n)**: Averages L_n and R_n, approximating the area with trapezoids.

$$T_n = \frac{\Delta x}{2}[f(x_0) + 2f(x_1) + 2f(x_2) + \cdots + 2f(x_{n-1}) + f(x_n)]$$

4. **Simpson's Rule (S_n)**: Approximates the curve using parabolas (n must be even).

$$S_n = \frac{\Delta x}{3}[f(x_0) + 4f(x_1) + 2f(x_2) + 4f(x_3) + \cdots + 2f(x_{n-2}) + 4f(x_{n-1}) + f(x_n)]$$

1.2 Polar Coordinates

A point in the plane can be represented by polar coordinates (r, θ), where r is the distance from the origin (pole) and θ is the angle measured counterclockwise from the positive x-axis (polar axis).

- **Conversion to Cartesian**: $x = r\cos\theta$, $y = r\sin\theta$.
- **Conversion to Polar**: $r^2 = x^2 + y^2$, $\tan\theta = \frac{y}{x}$.

Area Enclosed by a Polar Curve
The area A of the region bounded by a polar curve $r = f(\theta)$ and the rays $\theta = \alpha$ and $\theta = \beta$ is:

$$A = \int_\alpha^\beta \frac{1}{2}[f(\theta)]^2\,d\theta = \int_\alpha^\beta \frac{1}{2}r^2\,d\theta$$

Tip: Exploit symmetry whenever possible to simplify the calculation. For example, if the curve is symmetric about the x-axis or y-axis, calculate the area in one quadrant and multiply by the appropriate factor.

Arc Length of a Polar Curve
The arc length L of a polar curve $r = f(\theta)$ from $\theta = \alpha$ to $\theta = \beta$ is:

$$L = \int_{\alpha}^{\beta} \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2}\, d\theta$$

2 Exercises

2.1 Numerical Integration

Exercise 2.1. Use L_4, R_4, T_4, and S_4 to evaluate the following integral:

$$\int_0^1 \frac{dx}{1+x^2} = \frac{1}{4}\pi$$

Solution. **Idea.** Identify $f(x) = \frac{1}{1+x^2}$, $a = 0$, $b = 1$, and $n = 4$. First, calculate Δx and the values of x_i and $f(x_i)$.

We have $\Delta x = \frac{1-0}{4} = \frac{1}{4}$. The partition points are $x_i = 0, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}, 1$. Let's construct a table of values for $f(x_i)$:

i	0	1	2	3	4
x_i	0	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{3}{4}$	1
$f(x_i)$	1	$\frac{16}{17}$	$\frac{4}{5}$	$\frac{16}{25}$	$\frac{1}{2}$

Now apply the formulas for each approximation method:

(1) Left Endpoint Approximation (L_4):

$$\begin{aligned} L_4 &= \Delta x[f(x_0) + f(x_1) + f(x_2) + f(x_3)] \\ &= \frac{1}{4}\left(1 + \frac{16}{17} + \frac{4}{5} + \frac{16}{25}\right) \\ &= \frac{1437}{1700} \approx 0.8453 \end{aligned}$$

(2) Right Endpoint Approximation (R_4):

$$\begin{aligned} R_4 &= \Delta x[f(x_1) + f(x_2) + f(x_3) + f(x_4)] \\ &= \frac{1}{4}\left(\frac{16}{17} + \frac{4}{5} + \frac{16}{25} + \frac{1}{2}\right) \\ &= \frac{2449}{3400} \approx 0.7203 \end{aligned}$$

(3) Trapezoidal Rule (T_4):

$$\begin{aligned} T_4 &= \frac{\Delta x}{2}[f(x_0) + 2f(x_1) + 2f(x_2) + 2f(x_3) + f(x_4)] \\ &= \frac{1}{8}\left(1 + 2\left(\frac{16}{17}\right) + 2\left(\frac{4}{5}\right) + 2\left(\frac{16}{25}\right) + \frac{1}{2}\right) \\ &= \frac{5323}{6800} \approx 0.7828 \end{aligned}$$

(4) Simpson's Rule (S_4):

$$\begin{aligned}
S_4 &= \frac{\Delta x}{3}[f(x_0)+4f(x_1)+2f(x_2)+4f(x_3)+f(x_4)] \\
&= \frac{1}{12}\left(1+4\left(\frac{16}{17}\right)+2\left(\frac{4}{5}\right)+4\left(\frac{16}{25}\right)+\frac{1}{2}\right) \\
&= \frac{8011}{10200} \approx 0.785392
\end{aligned}$$

(Note: The true value is $\frac{\pi}{4} \approx 0.785398$, so S_4 gives a very accurate approximation.) □

2.2 Polar Coordinate: Area Enclosed by Curve

Exercise 2.2. Draw the figure of the following curves and evaluate the areas enclosed by these curves.

1. $r = a\theta$, $\theta = 0$ to $\theta = 2\pi$
2. $r = ae^{\theta}$, $\theta = 0$ to $\theta = 2\pi$
3. $r^2 = a^2 \cos 2\theta$

Solution. **Idea.** Apply the polar area formula $A = \frac{1}{2}\int r^2\,d\theta$. For closed curves, consider symmetry to simplify the integral limits.

(1) Archimedean spiral ($r = a\theta$)
The area swept out as θ goes from 0 to 2π is:

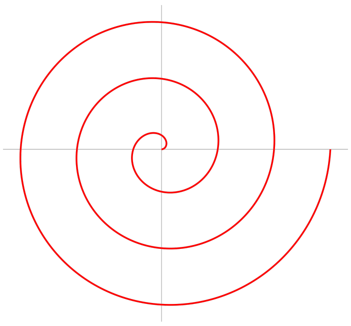
$$\begin{aligned}
A &= \int_0^{2\pi} \frac{1}{2} r^2\,d\theta \\
&= \frac{1}{2}\int_0^{2\pi} (a\theta)^2\,d\theta \\
&= \frac{a^2}{2}\int_0^{2\pi} \theta^2\,d\theta \\
&= \frac{a^2}{2}\left[\frac{\theta^3}{3}\right]_0^{2\pi} \\
&= \frac{a^2}{2}\left(\frac{8\pi^3}{3} - 0\right) \\
&= \frac{4}{3}\pi^3 a^2
\end{aligned}$$

(2) Logarithmic spiral ($r = ae^{\theta}$)
The area swept out from $\theta = 0$ to $\theta = 2\pi$ is:

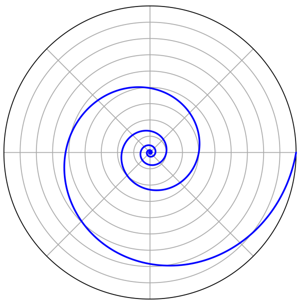
$$\begin{aligned}
A &= \int_0^{2\pi} \frac{1}{2} r^2\,d\theta \\
&= \frac{1}{2}\int_0^{2\pi} (ae^{\theta})^2\,d\theta \\
&= \frac{a^2}{2}\int_0^{2\pi} e^{2\theta}\,d\theta \\
&= \frac{a^2}{2}\left[\frac{1}{2}e^{2\theta}\right]_0^{2\pi} \\
&= \frac{a^2}{4}(e^{4\pi} - 1)
\end{aligned}$$

(3) Lemniscate of Bernoulli ($r^2 = a^2 \cos 2\theta$)
Notice that r^2 must be non-negative, so $\cos 2\theta \geq 0$. This means for the principal loop, $2\theta \in [-\frac{\pi}{2}, \frac{\pi}{2}]$, or $\theta \in [-\frac{\pi}{4}, \frac{\pi}{4}]$. The period of $\cos 2\theta$ is π. Moreover, we have $\cos 2(-\theta) = \cos 2\theta$ and $\cos 2(\pi - \theta) = \cos(2\pi - 2\theta) = \cos 2\theta$, which shows the curve is symmetric with respect to both the x-axis and the y-axis. Therefore, we only need to calculate the area in the first quadrant, where $\theta \in [0, \frac{\pi}{4}]$, and then multiply the result by 4.

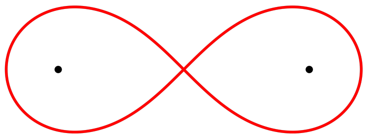
$$\begin{aligned} A &= 4 \times \int_0^{\frac{\pi}{4}} \frac{1}{2} r^2 \, d\theta \\ &= 2 \int_0^{\frac{\pi}{4}} a^2 \cos 2\theta \, d\theta \\ &= 2a^2 \left[\frac{1}{2} \sin 2\theta \right]_0^{\frac{\pi}{4}} \\ &= a^2 \left(\sin \frac{\pi}{2} - \sin 0 \right) \\ &= a^2 \end{aligned}$$



Archimedean spiral



Logarithmic spiral



Lemniscate of Bernoulli

□

2.3 Polar Coordinate: Arc Length

Exercise 2.3. Find the arc length of the following curves.

1. $r = a(1 - \cos \theta), \ 0 \leq \theta \leq 2\pi$
2. $r = a\theta, \ 0 \leq \theta \leq 2\pi$
3. $r = a \sin^3 \frac{\theta}{3}, \ 0 \leq \theta \leq 3\pi$

Solution. **Idea.** We use the arc length formula $L = \int \sqrt{r^2 + (r')^2} \, d\theta$. It is highly recommended to compute and simplify $r^2 + (r')^2$ first before placing it inside the square root.

(1) For $r = a(1-\cos\theta)$, we have $r' = a\sin\theta$. First, calculate $r^2 + (r')^2$:

$$\begin{aligned} r^2 + (r')^2 &= a^2(1-\cos\theta)^2 + (a\sin\theta)^2 \\ &= a^2(1 - 2\cos\theta + \cos^2\theta) + a^2\sin^2\theta \\ &= a^2(1 - 2\cos\theta + \cos^2\theta + \sin^2\theta) \\ &= a^2(2 - 2\cos\theta) \\ &= 2a^2(1-\cos\theta) \end{aligned}$$

Using the half-angle formula $1 - \cos\theta = 2\sin^2\frac{\theta}{2}$, we get:

$$r^2 + (r')^2 = 2a^2\left(2\sin^2\frac{\theta}{2}\right) = 4a^2\sin^2\frac{\theta}{2}$$

Since $0 \le \theta \le 2\pi$, we have $0 \le \frac{\theta}{2} \le \pi$, meaning $\sin\frac{\theta}{2} \ge 0$. Thus, the square root simplifies nicely:

$$\sqrt{r^2 + (r')^2} = 2a\sin\frac{\theta}{2}$$

Now evaluate the integral:

$$\begin{aligned} L &= \int_0^{2\pi} 2a\sin\frac{\theta}{2}\,d\theta \\ &= 2a\left[-2\cos\frac{\theta}{2}\right]_0^{2\pi} \\ &= -4a(\cos\pi - \cos 0) \\ &= -4a(-1-1) = 8a \end{aligned}$$

(2) For $r = a\theta$, we have $r' = a$. Calculate $r^2 + (r')^2$:

$$r^2 + (r')^2 = a^2\theta^2 + a^2 = a^2(1+\theta^2)$$

Thus, the arc length integral is:

$$L = \int_0^{2\pi} \sqrt{a^2(1+\theta^2)}\,d\theta = \int_0^{2\pi} a\sqrt{1+\theta^2}\,d\theta$$

We can evaluate this integral using the known formula for $\int \sqrt{1+x^2}\,dx$ (derived via trigonometric substitution $x = \tan t$):

$$\begin{aligned} L &= a\left[\frac{\theta}{2}\sqrt{1+\theta^2} + \frac{1}{2}\ln(\theta + \sqrt{1+\theta^2})\right]_0^{2\pi} \\ &= a\left(\frac{2\pi}{2}\sqrt{1+4\pi^2} + \frac{1}{2}\ln(2\pi + \sqrt{1+4\pi^2}) - 0\right) \\ &= \pi a\sqrt{1+4\pi^2} + \frac{a}{2}\ln(2\pi + \sqrt{1+4\pi^2}) \end{aligned}$$

(3) For $r = a\sin^3\frac{\theta}{3}$, using the chain rule we have $r' = a \cdot 3\sin^2\frac{\theta}{3} \cdot \cos\frac{\theta}{3} \cdot \frac{1}{3} = a\sin^2\frac{\theta}{3}\cos\frac{\theta}{3}$. Calculate $r^2 + (r')^2$:

$$\begin{aligned} r^2 + (r')^2 &= \left(a\sin^3\frac{\theta}{3}\right)^2 + \left(a\sin^2\frac{\theta}{3}\cos\frac{\theta}{3}\right)^2 \\ &= a^2\sin^6\frac{\theta}{3} + a^2\sin^4\frac{\theta}{3}\cos^2\frac{\theta}{3} \\ &= a^2\sin^4\frac{\theta}{3}\left(\sin^2\frac{\theta}{3} + \cos^2\frac{\theta}{3}\right) \\ &= a^2\sin^4\frac{\theta}{3} \end{aligned}$$

Taking the square root (since $0 \leq \theta \leq 3\pi$, we have $0 \leq \frac{\theta}{3} \leq \pi$, so $\sin \frac{\theta}{3} \geq 0$):

$$\sqrt{r^2 + (r')^2} = a \sin^2 \frac{\theta}{3}$$

Now integrate using the half-angle identity $\sin^2 x = \frac{1-\cos 2x}{2}$:

$$\begin{aligned}
L &= \int_0^{3\pi} a \sin^2 \frac{\theta}{3}\, d\theta \\
&= a \int_0^{3\pi} \frac{1 - \cos \frac{2\theta}{3}}{2}\, d\theta \\
&= \frac{a}{2} \left[\theta - \frac{3}{2} \sin \frac{2\theta}{3} \right]_0^{3\pi} \\
&= \frac{a}{2} \left(3\pi - \frac{3}{2} \sin(2\pi) - (0 - 0) \right) \\
&= \frac{3\pi a}{2}
\end{aligned}$$

$\square$